

RetailPulse Inc.

Sales Performance Analytics
End-to-End Business Analysis Project

Prepared by

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Tools Used: SQL (SQLite) • Tableau • Microsoft Word

Dashboard: public.tableau.com/app/profile/aastha.baria

March 2024

Project Overview

RetailPulse Inc. is a fictional mid-size retail chain operating 42 stores across 5 Canadian regions — Ontario, Quebec, BC, Alberta, and Atlantic — with an e-commerce channel launched in 2022. Despite opening 6 new stores, the company grew revenue by only 2.3% last year. Leadership lacked a centralized view of performance, and decisions were being made from disconnected regional spreadsheets.

This project covers the full BA lifecycle across 4 phases: Business Requirements Document, SQL Query Portfolio, Data Analysis, and an Interactive Tableau Dashboard.

Project Phases at a Glance

- **Phase 1 — Business Requirements Document:** Stakeholder analysis, elicitation techniques, functional & non-functional requirements, acceptance criteria
- **Phase 2 — SQL Query Portfolio:** 7+ queries on a 5-table relational database covering JOINS, aggregations, CASE WHEN, window functions, and CTEs
- **Phase 3 — Data Analysis:** Sales performance analysis across 5 regions, departments, and channels
- **Phase 4 — Tableau Dashboard:** Published interactive dashboard — revenue by region, channel split, top departments

Phase 1 — Business Requirements Document (BRD)

Business Problem Statement

RetailPulse Inc. is currently lacking a standardized system for recording and tracking sales performance. Sales data is maintained across disconnected regional spreadsheets spanning 42 stores, 5 regions, and an e-commerce channel, making it difficult for the leadership team to get a consistent and timely view of results. As a result, the business grew revenue by only 2.3% last year despite opening 6 new stores. Regional managers spend considerable time compiling manual reports rather than acting on insights. Without a centralized solution, the company risks continued margin erosion and misallocation of investment.

Stakeholder Register

- **Sarah Chen** (VP of Sales (Project Sponsor)) — Influence: High. Needs actionable insights for executive decisions
- **David Okafor** (IT Director) — Influence: High. Concerned about integration, data loss, and 5-year sustainability
- **Marcus Reid** (Regional Sales Manager, ON) — Influence: Medium. Needs on-the-go store access, simple tool
- **Priya Nair** (Head of E-commerce) — Influence: Medium. Wants e-comm data on equal footing with in-store
- **Linda Tran** (Finance Analyst) — Influence: Low. Needs accurate P&L; data, dashboard must match source systems

Elicitation Techniques

- **1-on-1 Interviews:** Conducted with all 5 stakeholders to uncover individual concerns and expectations without group influence.
- **Requirements Workshop:** Facilitated with Marcus Reid, Priya Nair, David Okafor, and store managers from the 2 lowest-performing regions. Key conflict surfaced: IT wanted simplicity, Sales wanted rich filtering.
- **Survey:** Distributed to all 42 stores. Results: 78% rated current tools inadequate, average 5 hours/week spent on manual reporting.
- **Document Analysis:** Reviewed existing spreadsheets. Found: duplicate records, missing store IDs and product categories, inconsistent date formats, no common identifier linking in-store and e-commerce data.

Key Requirements Summary

- FR-01: Display daily, WTD, and YTD sales figures per store and region
- FR-02: Sales performance broken down by department
- FR-03: Real-time data refresh
- FR-04: Filters by department, time period, and date range
- FR-05: Role-based access control
- FR-06: Executive summary view across all 42 stores, 5 regions, e-commerce
- FR-07: E-commerce data alongside in-store using consistent metrics
- FR-08: Automated alert when store falls below daily sales threshold
- FR-09: Exportable reports in Excel and PDF
- FR-10: Full audit log of user access and data exports

Key Non-Functional Requirements

- NFR-01: Dashboard loads within 5-10 seconds
- NFR-02: Minimum 99.5% uptime with automated failover
- NFR-03: IT team restores system within 4 hours of outage
- NFR-04: Secure login with role-based permissions

- NFR-05: Accessible on desktop and mobile browsers

Acceptance Criteria

- AC-01: Regional manager sees real-time sales filterable by dept, date, and store on login
- AC-02: Dashboard shows whether each store met/missed daily budget target with trend analysis
- AC-03: Dashboard figures match source data within 15-minute refresh window
- AC-04: UAT confirms each role only sees data assigned to that role
- AC-05: Load test confirms dashboard loads within 10 seconds with simultaneous users

Phase 2 — SQL Query Portfolio

A 5-table relational database was designed for RetailPulse: stores, products, customers, sales_transactions, and sales_targets. All queries were written in SQLite and tested against 120 real transactions spanning 2022 and 2023.

Q1 — Total Revenue by Region

Business question: Which of the 5 regions generated the most revenue overall?

The screenshot shows a SQL query editor interface. On the left, a sidebar lists database tables: customers, demo, products, sales_targets, sales_transactions, and stores. The main editor displays a SQL query to calculate total revenue by region. The query is as follows:

```
1 SELECT
2   s.region,
3   SUM(t.total_amount) AS total_revenue
4 FROM sales_transactions t
5 JOIN stores s ON t.store_id = s.store_id
6 GROUP BY s.region
7 ORDER BY total_revenue DESC;
```

Below the query, the results are displayed in a table:

region	total_revenue
Ontario	9690.5
Quebec	4723.75
BC	4328.5
Alberta	1695
Atlantic	875

On the right, a 'History' panel shows previous queries and errors. Two failed queries are visible, both with syntax errors related to the 'FROM' and 'GROUP' clauses. The first error message is: 'SQLITE_ERROR: sqlite3 result code 1: near "FROM": syntax error'. The second error message is: 'SQLITE_ERROR: sqlite3 result code 1: near "GROUP": syntax error'. Both error messages include a hint: 'You have reached the hint limit in your SQL code according to your plan.' and an 'AI Help' button.

Result: Ontario led all regions with \$9,690.50 in total revenue.

Q3 — Monthly Sales Trend (2023)

Business question: Which months had the highest and lowest revenue in 2023?

The screenshot shows the nrisory SQL editor interface. On the left, a sidebar lists databases: Private.DB, Demo.Memory, SQLite (selected), DuckDB, and PGLite. The main editor displays a SQL query to find the strongest and weakest months by total revenue. The query is as follows:

```
1 SELECT
2   strftime('%m', transaction_date) AS month,
3   SUM(total_amount) AS total_revenue
4 FROM sales_transactions
5 WHERE strftime('%Y', transaction_date) = '2023'
6 GROUP BY strftime('%m', transaction_date)
7 ORDER BY month ASC;
```

The results table shows the following data:

month	total_revenue
01	877.25
02	947.5
03	828
04	1107.25
05	1000.5
06	1049
07	1046.5
08	728
09	1207.75
10	1001
11	1270
12	1776.5

On the right, the history panel shows the executed query and a message: "SQLITE_ERROR: sqlite3 result code 1: near 'FROM': syntax error". Below this, a hint message states: "You have reached the hint limit in your SQL code according to your plan." and "AI Help".

Result: December was the strongest month (\$1,776.50). August was the weakest (\$728).

Q4 — Store Performance vs Monthly Target

Business question: Did each store hit or miss its monthly sales target?

The screenshot shows the nrisory SQL editor interface. On the left, a sidebar lists databases: Demo.Memory, SQLite (selected), DuckDB, PGLite, MariaDB, and PostgreSQL. The main editor displays a SQL query to check if each store hit or missed its monthly sales target. The query is as follows:

```
1 SELECT
2   s.store_name,
3   st.month,
4   SUM(t.total_amount) AS actual_revenue,
5   st.target_amount,
6   CASE WHEN SUM(t.total_amount) >= st.target_amount
7     THEN 'Hit'
8     ELSE 'Missed'
9   END AS status
10 FROM sales_transactions t
11 JOIN stores s ON t.store_id = s.store_id
12 JOIN sales_targets st ON t.store_id = st.store_id
13 AND CAST(strftime('%m', t.transaction_date) AS INTEGER) = st.month
```

The results table shows the following data:

store_name	month	actual_revenue	target_amount	status
BC-Vancouver-Downtown	1	162	80000	Missed
ON-Toronto-Downtown	1	330.25	85000	Missed
ON-Toronto-North	1	165	70000	Missed
QC-Montreal-Centre	1	220	78000	Missed
AB-Calgary-Centre	2	140	68000	Missed
ON-Toronto-Downtown	2	464	80000	Missed
QC-Montreal-Centre	2	218.5	74000	Missed
BC-Vancouver-Downtown	3	170	84000	Missed
ON-Toronto-Downtown	3	342	90000	Missed
ON-Toronto-North	3	100	75000	Missed
QC-Montreal-Centre	3	216	82000	Missed
AB-Calgary-Centre	4	210	74000	Missed
BC-Vancouver-Downtown	4	306	82000	Missed

On the right, the history panel shows the executed query and a message: "SQLITE_ERROR: sqlite3 result code 1: near 'FROM': syntax error". Below this, a hint message states: "You have reached the hint limit in your SQL code according to your plan." and "AI Help".

Result: Most stores missed their targets — highlighting the revenue growth problem identified in the BRD.

Q5 — In-Store vs Online Revenue

Business question: How does e-commerce revenue compare to physical store revenue?

SQLite

```
1 SELECT channel,
2 SUM(total_amount) AS total_revenue,
3 COUNT(transaction_id) AS num_transactions
4 FROM sales_transactions
5 GROUP BY channel
```

channel	total_revenue	num_transactions
in-store	17329.75	102
online	3983	18

SQLite

```
SELECT channel,
SUM(total_amount) AS total_revenue,
COUNT(transaction_id) AS num_transac
FROM s
...
```

01:22:15

SQLite

```
SELECT
s.store_name,
st.month,
SUM(t.total_amount) AS actual_re
st.target_amo
...
```

01:21:41

SQLite

```
SELECT s.store_name, st.month,
SUM(t.total_amount) AS actual,
st.target_amount,
CASE WHEN SUM(total_
...
```

SQLite_ERROR: sqlite3 result code 1: near "FROM": syntax error

You have reached the hint limit in your SQL code according to your plan.

AI Help

01:20:38

Result: In-store revenue (\$17,329.75) significantly outperformed online (\$3,983) — 102 vs 18 transactions.

Q6 — Revenue by Department and Region

Business question: Which departments perform best in which regions? (3-table JOIN)

The screenshot shows a SQL IDE interface with a query editor, a table viewer, and a console. The query is a 3-table JOIN:

```
1 SELECT
2   s.region,
3   p.department,
4   SUM(t.total_amount) AS total_revenue
5 FROM sales_transactions t
6 JOIN stores s ON t.store_id = s.store_id
7 JOIN products p ON t.product_id = p.product_id
8 GROUP BY s.region, p.department
9 ORDER BY s.region DESC, total_revenue DESC;
```

The table viewer displays the results of the query:

region	department	total_revenue
Quebec	Apparel	2067.75
Quebec	Electronics	1386
Quebec	Home & Kitchen	1080
Quebec	Health & Beauty	190
Ontario	Apparel	5332.5
Ontario	Electronics	1899
Ontario	Home & Kitchen	1527
Ontario	Toys & Games	455
Ontario	Health & Beauty	327
BC	Apparel	1591.5
BC	Home & Kitchen	1404
BC	Electronics	1123

The console shows an error message:

```
SQLITE_ERROR: sqlite3 result code 1: no such column: sales_targets
```

You have reached the hint limit in your SQL code according to your plan.

Result: Apparel led in all regions. Ontario had the strongest overall department performance.

Q12 — Year-over-Year Revenue Growth by Region (CTE)

Business question: Which regions grew from 2022 to 2023 and by how much?

The screenshot shows a SQL IDE interface with a query editor, a table viewer, and a console. The query uses a CTE to calculate year-over-year revenue growth:

```
8   GROUP BY s.region, year
9 )
10 SELECT
11   y1.region,
12   y1.revenue AS revenue_2022,
13   y2.revenue AS revenue_2023,
14   ROUND((y2.revenue - y1.revenue) / y1.revenue * 100, 2) AS growth_pct
15 FROM yearly y1
16 JOIN yearly y2 ON y1.region = y2.region
17 WHERE y1.year = 2022
18   AND y2.year = 2023
19 ORDER BY growth_pct DESC;
```

The table viewer displays the results of the query:

region	revenue_2022	revenue_2023	growth_pct
Quebec	1764.5	2959.25	67.71
BC	1743.75	2584.75	48.23
Ontario	3965.25	5725.25	44.39
Alberta	660	1035	0
Atlantic	340	535	0

The console shows an error message:

```
SQLITE_ERROR: sqlite3 result code 1: near "FROM": syntax error
```

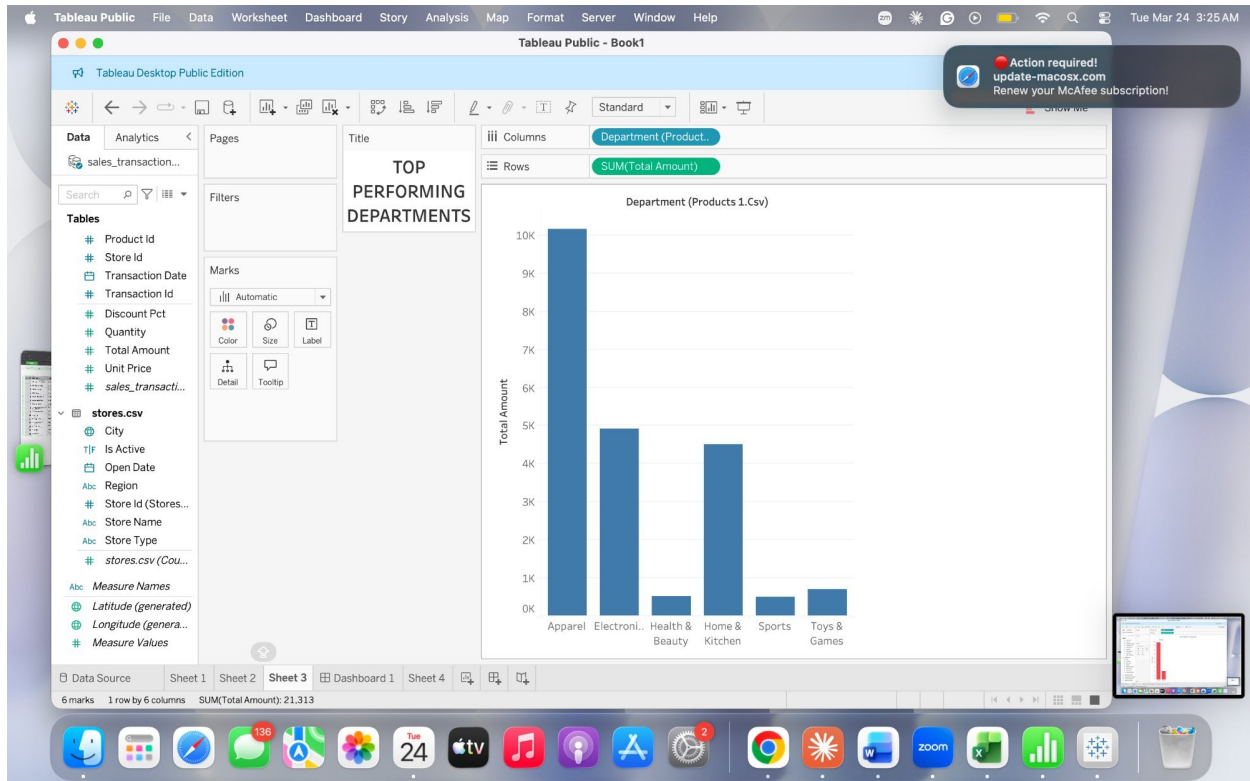
You have reached the hint limit in your SQL code according to your plan.

Result: Quebec grew 67.71%, BC grew 48.23%, Ontario grew 44.39%. Alberta and Atlantic showed 0% growth due to limited 2022 data.

Phase 3 & 4 — Data Analysis & Tableau Dashboard

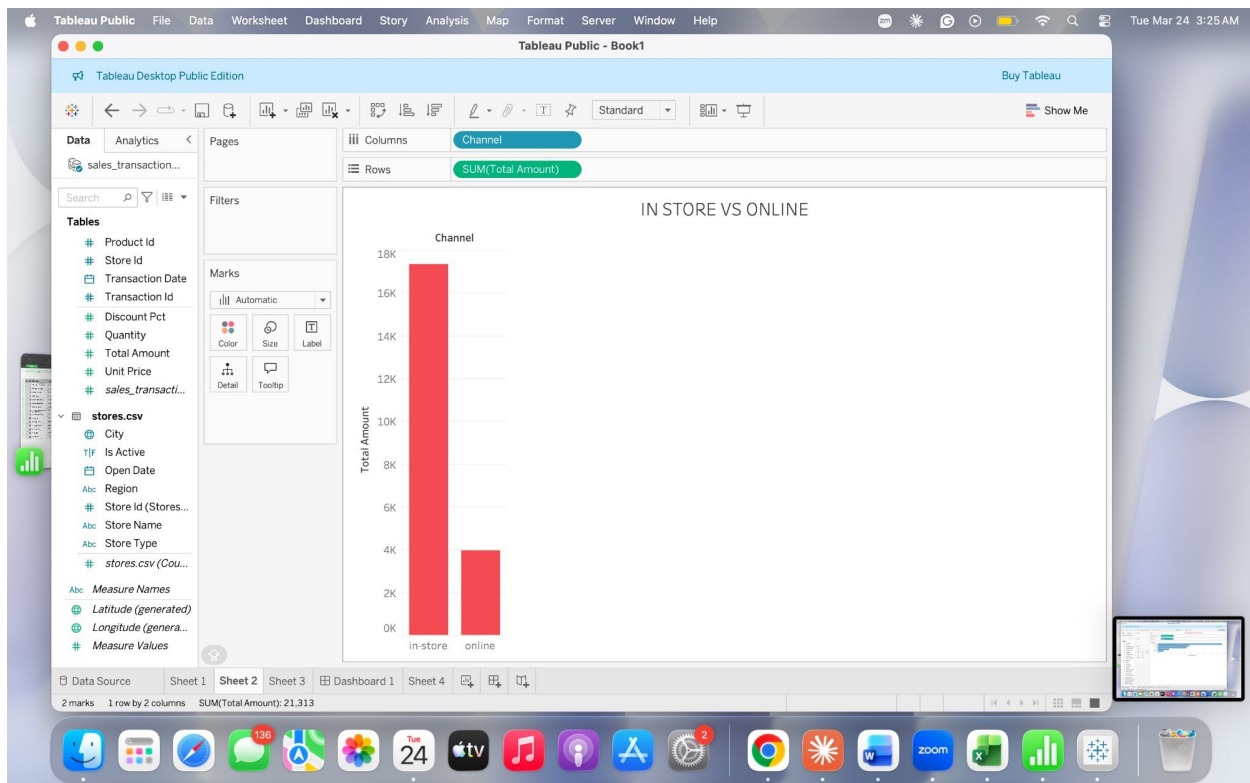
Sales data was loaded into Tableau Public and analyzed across regions, departments, and channels. An interactive dashboard was built and published combining all key insights into one view.

Revenue by Region



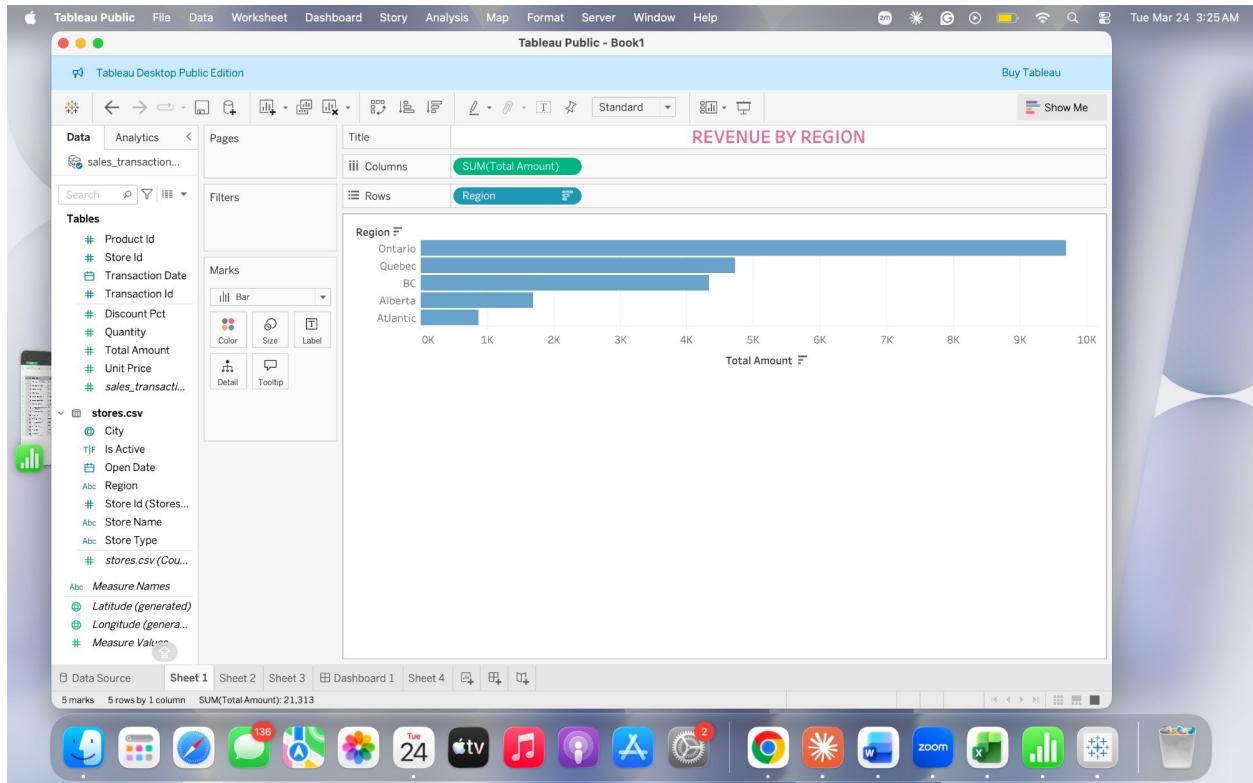
Ontario dominated regional revenue, followed by Quebec and BC. Atlantic showed the lowest performance.

In-Store vs Online Channel Split



In-store sales (~\$17K) far outpaced online (~\$4K), confirming e-commerce is still an underutilized channel.

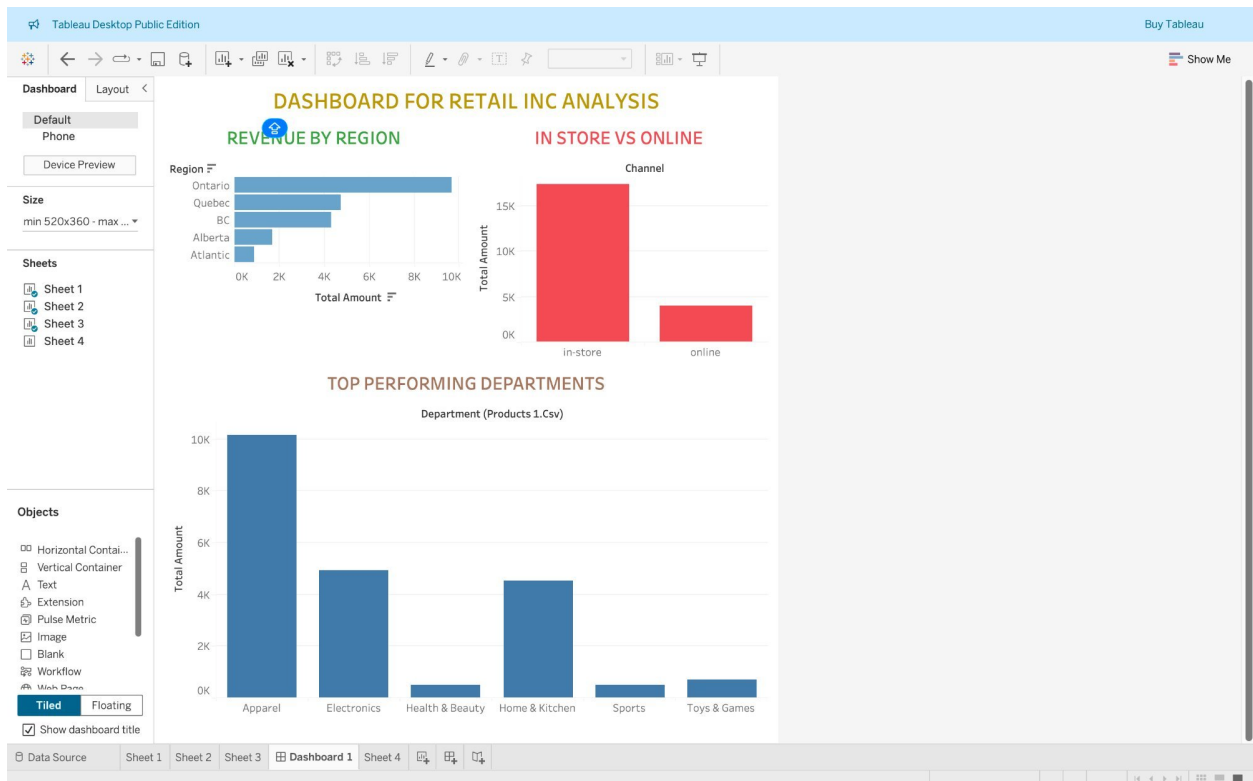
Top Performing Departments



Apparel was the clear leader at ~\$10K, followed by Electronics and Home & Kitchen. Sports and Toys & Games underperformed.

Final Dashboard — All Charts Combined

All three visualizations were combined into a single Tableau Public dashboard titled 'Dashboard for RetailPulse Analysis'. The dashboard is published and publicly accessible.



Live dashboard: public.tableau.com/app/profile/aastha.baria/viz/Book1_17743377719870/Dashboard1

Key Insights & Findings

- Ontario is the strongest performing region — contributing the highest revenue across both 2022 and 2023.
- Quebec showed the highest year-over-year growth at 67.71%, suggesting strong momentum.
- Apparel is the top department by revenue across all regions, while Sports and Toys & Games significantly underperform.
- In-store sales account for roughly 81% of total revenue — e-commerce at 19% is an area of opportunity.
- Nearly all stores missed their monthly sales targets in 2023, confirming the 2.3% growth gap identified in the problem statement.
- Document analysis during elicitation revealed duplicate records, missing fields, and no common key linking in-store and e-commerce data — root causes of the reporting problem.

Tools & Technologies

- SQL (SQLite) — Query writing, data extraction, window functions, CTEs
- Tableau Public — Data visualization, interactive dashboard, published online
- Microsoft Word — BRD documentation, professional formatting
- sqliteonline.com — Online SQL environment for query testing

About the Author

Aastha Baria is an ECBA-certified Business Analyst with a Post-Graduate Certificate in Business Insights and Analytics from Humber College and close to 2 years of experience as a Department Manager at Walmart Canada. This project was independently designed and completed to demonstrate end-to-end BA skills across requirements gathering, SQL data analysis, and data visualization.

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